LAB REPORT 7

MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

GROUP 6

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Abstract

This experiment demonstrates the practical use of Bluetooth communication for wireless environmental monitoring and simple remote control using an ESP32 microcontroller. A DHT11 temperature and humidity sensor was used to collect real-time environmental data, which the ESP32 transmitted wirelessly to a smartphone through its built-in Bluetooth Serial functionality. In addition to monitoring, an LED was integrated into the system to act as a basic actuator or simulated FAN indicator. The LED's state could be controlled remotely by sending simple text commands such as "1" and "0" from a Bluetooth terminal application running on the smartphone.

The overall goal of the experiment was to explore how sensing, microcontroller programming, and wireless communication can be combined to create a simple yet functional mechatronics system. The results showed that the ESP32 was able to accurately read and transmit sensor data, respond immediately to Bluetooth commands, and control the LED without delays, confirming the effectiveness of Bluetooth for short-range wireless control. This experiment highlights fundamental concepts used in many modern applications, including home automation, IoT devices, and remote monitoring systems.

***Keywords*:** ESP32, Bluetooth, DHT11, Wireless Monitoring, LED Control

Table of Contents

Abstract.......... 1
1. Introduction.......... 3
2. Materials and Equipments.......... 4
3. Experimental Setup.......... 5
4. Methodology.......... 6
5. Data Collection.......... 10
6. Data Analysis.......... 11
7. Results.......... 12
8. Discussion.......... 13
9. Conclusion.......... 14
10. Recommendations.......... 15
11. References.......... 16
Appendix A - Circuit Diagram.......... 17
Appendix B - Coding Snippets.......... 18
Appendix C - Results.......... 20
Acknowledgments.......... 21

WEEK 8: REMOTE TEMPERATURE MONITORING AND CONTROL VIA BLUETOOTH

1. Introduction

Bluetooth is a wireless communication technology that allows devices to exchange data over short distances without using cables. It is widely used in portable electronics, home automation, and mechatronics systems because it is simple, low-cost, and easy to integrate. In this experiment, Bluetooth is used to send sensor readings from the ESP32 microcontroller to a smartphone and to receive commands for remote control.

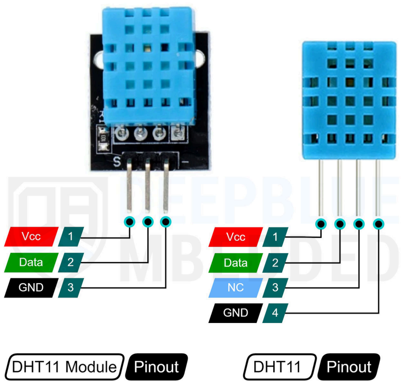

Figure 1.0 Shows the DHT 11 module and DHT 11 sensor Pin

The **DHT11 sensor** is a basic digital sensor used to measure **temperature and humidity**. It is easy to use with microcontrollers like the ESP32 and provides real-time environmental data. In this system, the ESP32 reads the temperature and humidity from the DHT11 sensor and transmits the data to a paired smartphone using Bluetooth.

Additionally, the system includes an LED as a simple actuator, which can be turned ON or OFF by sending commands from the smartphone, simulating a FAN or indicator. This setup demonstrates the integration of sensors, microcontroller programming, and wireless communication in a basic mechatronics system. It also shows how a smartphone can function as both a monitoring device and a remote control interface.

Our main objectives in this experiment were:

- Measure temperature and humidity using a DHT11 sensor.
- Transmit sensor readings wirelessly to a smartphone via Bluetooth.
- Receive Bluetooth commands from the smartphone and control LED
- Demonstrate basic integration of sensors, microcontroller programming, and wireless communication.

2. Materials and Equipments

ESP32 Development Board	1
DHT11 Temperature and Humidity Sensor	1
Light Emitting Diode, LED (Indicator for FAN ON/OFF)	1
Resistor 220Ω	1
Smartphone with Serial Bluetooth Terminal app	1
Breadboard	
Jumper Wires	

3. Experimental Setup

The hardware setup was designed to integrate the ESP32, DHT11 sensor, and LED. The ESP32 was placed on a breadboard to simplify connections. The DHT11 sensor was connected to GPIO 21 for data output, and the LED was connected to GPIO 2. Power was supplied through the USB cable from a laptop.

Once the ESP32 was powered, it initialized Bluetooth communication and sensor readings. The smartphone was paired with the ESP32 under the device name "ESP32-FaZim's Logic" The Serial Bluetooth Terminal app on the smartphone displayed real-time readings from the DHT11 sensor, showing both temperature and humidity. The app also allowed the user to send simple commands like "LED ON" and "LED OFF" to control the LED.

The experiment setup ensured that data could be read and transmitted continuously while the ESP32 also listened for incoming commands. This setup demonstrates a basic closed-loop system where information is collected, transmitted, and acted upon remotely. The circuit setup is shown in Figure 3.0 below.

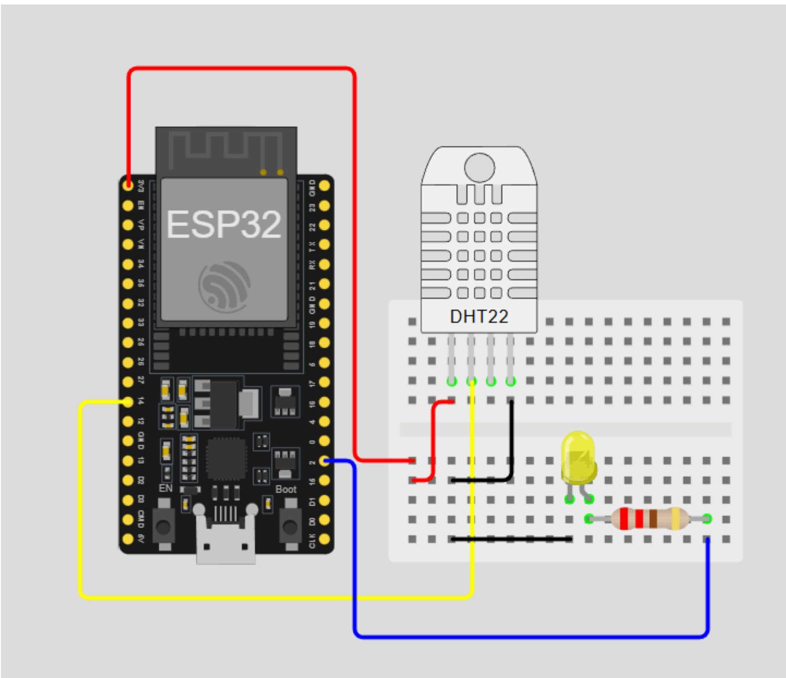

Figure 3.0 Shows the schematic of hardware setup using WokWi

4. Methodology

The methodology for this experiment involved four main steps: hardware setup, programming the ESP32, establishing Bluetooth communication, and testing the system. Each step ensured proper integration of sensors, microcontroller, and wireless communication for real-time monitoring and control.

4.1. Hardware Setup

The hardware setup focused on connecting the ESP32 microcontroller with the DHT11 sensor and the LED. The following connections were made:

DHT11 sensor	DATA pin → GPIO 14
	VCC → 3.3V
	GND → GND
LED:	Anode → GPIO 2
	Cathode → GND through a 220Ω resistor

Table 4.1 Shows the connection for the DHT11 sensor and LED pin.

All components were securely placed on a breadboard to ensure stable connections during testing. This setup allowed the ESP32 to read sensor values and control the LED while being powered through a USB cable from a laptop.

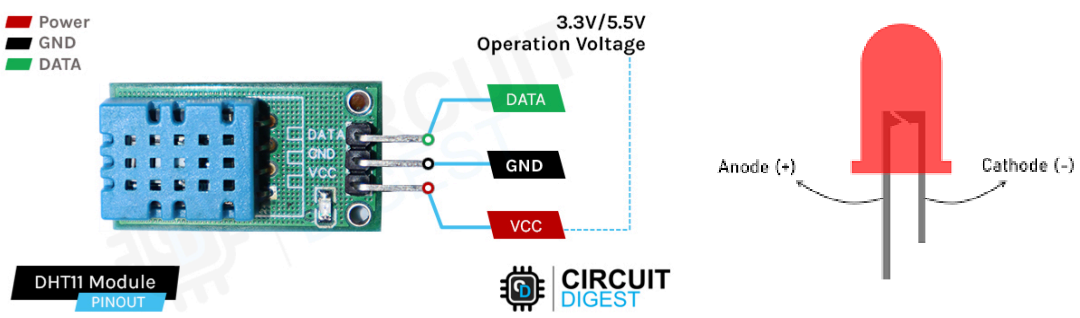

Figure 4.1 Shows the configuration pin for DHT11 and LED.

4.2. Programming the ESP32 through ArduinoIDE

```
#include <BluetoothSerial.h>
#include <DHT.h>

#define DHT11_PIN 21
#define DHTTYPE   DHT11
#define LED_PIN   2   // GPIO2 for external LED

String device_name = "ESP32_FaZim'sLogic";

BluetoothSerial SerialBT;
DHT dht11(DHT11_PIN, DHTTYPE);

#if !defined(CONFIG_BT_ENABLED) ||
!defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth not enabled in this ESP32
#endif

void setup() {
  Serial.begin(9600);
  SerialBT.begin(device_name);

  dht11.begin();

  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.println("ESP32 ready. Pair via Bluetooth.");
}

void loop() {
  // ---- Bluetooth LED Control ----
  if (SerialBT.available()) {
    char cmd = SerialBT.read();

    if (cmd == '1') {
      digitalWrite(LED_PIN, HIGH);
      SerialBT.println("LED ON");
    }
    else if (cmd == '0') {
      digitalWrite(LED_PIN, LOW);
```

```
      SerialBT.println("LED OFF");
    }
  }

  // ---- DHT11 readings ----
  float humi  = dht11.readHumidity();
  float tempC = dht11.readTemperature();
  float tempF = dht11.readTemperature(true);

  if (!isnan(humi) && !isnan(tempC)) {
    SerialBT.print("Humidity: ");
    SerialBT.print(humi);
    SerialBT.print("%  |  Temp: ");
    SerialBT.print(tempC);
    SerialBT.print("°C / ");
    SerialBT.print(tempF);
    SerialBT.println("°F");
  } else {
    SerialBT.println("Failed to read from DHT11");
  }

  delay(2000);
}
```

The ESP32 was programmed using the Arduino IDE. The program was designed to perform the following tasks:

- Initialize Bluetooth communication with the device name `"ESP32_FaZim'sLogic"`
- Read temperature and humidity from the DHT11 sensor every 1.5 seconds.
- Transmit real-time sensor data to the paired smartphone via Bluetooth.
- Continuously check for incoming Bluetooth commands from the smartphone.
- Control the LED based on the commands received (“1” or “0”).

This programming ensured that the ESP32 could function as both a sensor data transmitter and a remote-controlled actuator.

4.3. Bluetooth Connection

Once the ESP32 program was running, the smartphone was paired with the ESP32 using its Bluetooth device name. The **Serial Bluetooth Terminal app** was used to:

- Display live readings from the DHT11 sensor, including both temperature and humidity values.
- Send text commands to the ESP32 to control the LED ON and LED OFF.

This step verified the two-way communication capability between the microcontroller and the smartphone.

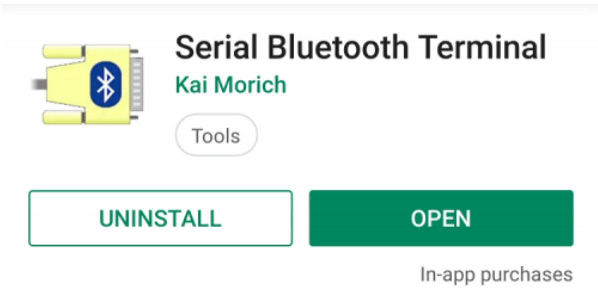

Figure 4.3 Shows the Serial Bluetooth Terminal apps in PlayStore

4.4. System Testing

After establishing hardware and software connections, the system was tested for proper functionality:

- Sensor readings were observed in real time on the smartphone.
- Commands were sent from the app to test the LED control.
- Stability, accuracy, and responsiveness of the system were checked to ensure reliable performance.
- Any errors in reading or command execution were noted and corrected as needed.

This step confirmed that the ESP32 could successfully integrate sensor readings, Bluetooth communication, and actuator control in a basic mechatronics system.

5. Data Collection

The system was tested over a period of 20–30 seconds to collect sample data from the DHT11 sensor and observe LED control. The readings were recorded at regular intervals, and the LED state was monitored to confirm proper response to commands sent from the smartphone. An example of the captured data is shown below:

Time, s	Temperature, °C	Humidity, %	LED
5	27.5	65	OFF
10	28.0	66	OFF
15	28.5	66	ON
20	28.5	67	ON

Table 5.0 Shows the observation of the LED

The data shows that the temperature **gradually increased** over the test period, reflecting environmental changes or slight warming from the hardware itself. The humidity remained relatively stable, which is expected for a short-duration indoor experiment.

The **LED responded** immediately to **commands** sent via the Bluetooth terminal app, demonstrating that the ESP32 successfully received and executed instructions in real time. This confirms the reliability and responsiveness of the Bluetooth communication link between the smartphone and the microcontroller.

Additionally, the continuous transmission of sensor data and the ability to control the LED simultaneously illustrate that the system can operate as a basic closed-loop mechatronics system, integrating sensing, wireless communication, and actuator control effectively.

6. Data Analysis

The collected data from the DHT11 sensor and LED responses were analyzed to evaluate the performance and reliability of the system. The temperature readings ranged between **27.5°C to 28.5°C**, which corresponds to normal indoor conditions. Humidity readings were stable between **65% to 67%**, showing that the DHT11 sensor is capable of providing consistent values over time.

The LED state changed immediately when commands were sent via Bluetooth, confirming that the ESP32 successfully received and executed remote commands. There were no delays noticeable to the human eye, which indicates that the Bluetooth communication is suitable for real-time monitoring and control in simple mechatronics applications.

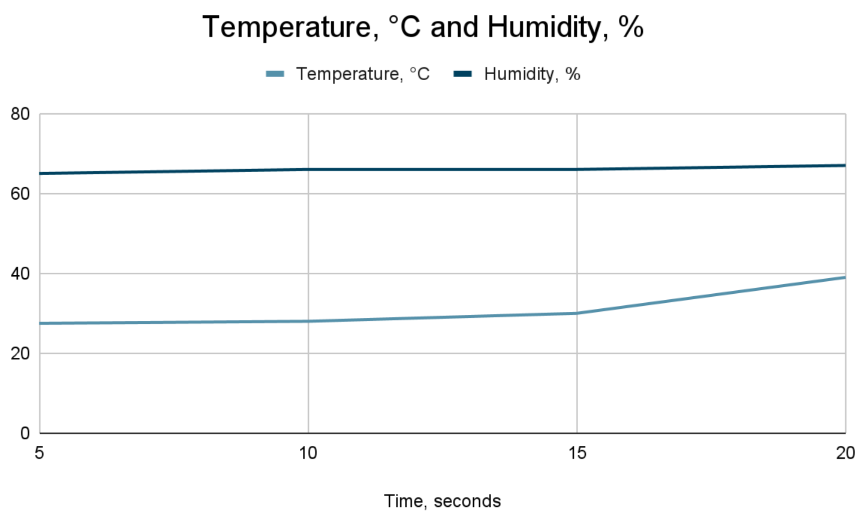

Figure 6.0 Shows the line graph based on the output fo the DHT11

From the data, it is clear that the system can provide:

1. **Real-time sensor monitoring** - Temperature and humidity are updated continuously on the smartphone.
2. **Remote control feedback** - LED reacts instantly to commands like “LED ON” and “LED OFF.”
3. **Reliable Bluetooth connection** - No command loss was observed during the experiment.

7. Results

The experiment successfully demonstrated a working Bluetooth-based monitoring and control system. The key results include:

1. **Temperature Monitoring:** The ESP32 successfully read temperature from the DHT11 sensor and transmitted it to the smartphone. Temperature readings were consistent and updated every 1.5 seconds.
2. **Humidity Monitoring:** Humidity readings were also stable and successfully transmitted via Bluetooth.
3. **LED Control:** The LED responded immediately to remote commands sent from the smartphone, showing that the ESP32 can act on user inputs in real time.
4. **Bluetooth Communication:** The ESP32 reliably maintained Bluetooth connection with the smartphone throughout the experiment. No disconnections were observed.
5. **System Integration:** The experiment demonstrated successful integration of a sensor (DHT11), actuator (LED), and wireless communication (Bluetooth) into a single system.

The results confirm that the system meets the experiment's objectives: real-time remote monitoring of temperature and humidity, and wireless control of an output device. The experiment also demonstrates basic principles of mechatronics, showing how sensors, microcontrollers, and wireless interfaces can work together.

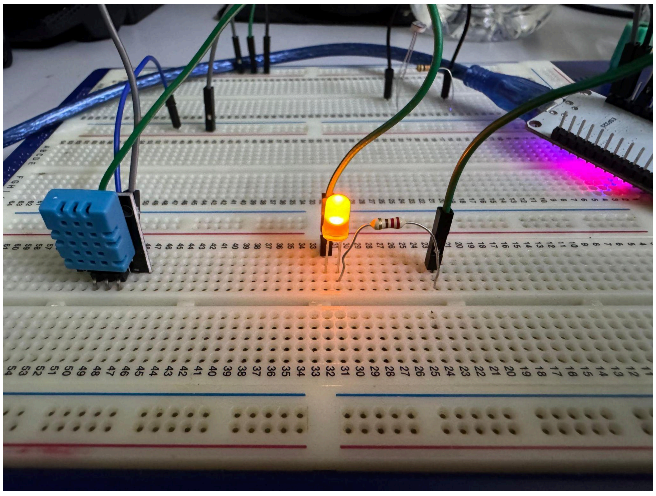

Figure 7.1 Shows the LED turn on after sending ‘1’ from Serial Bluetooth

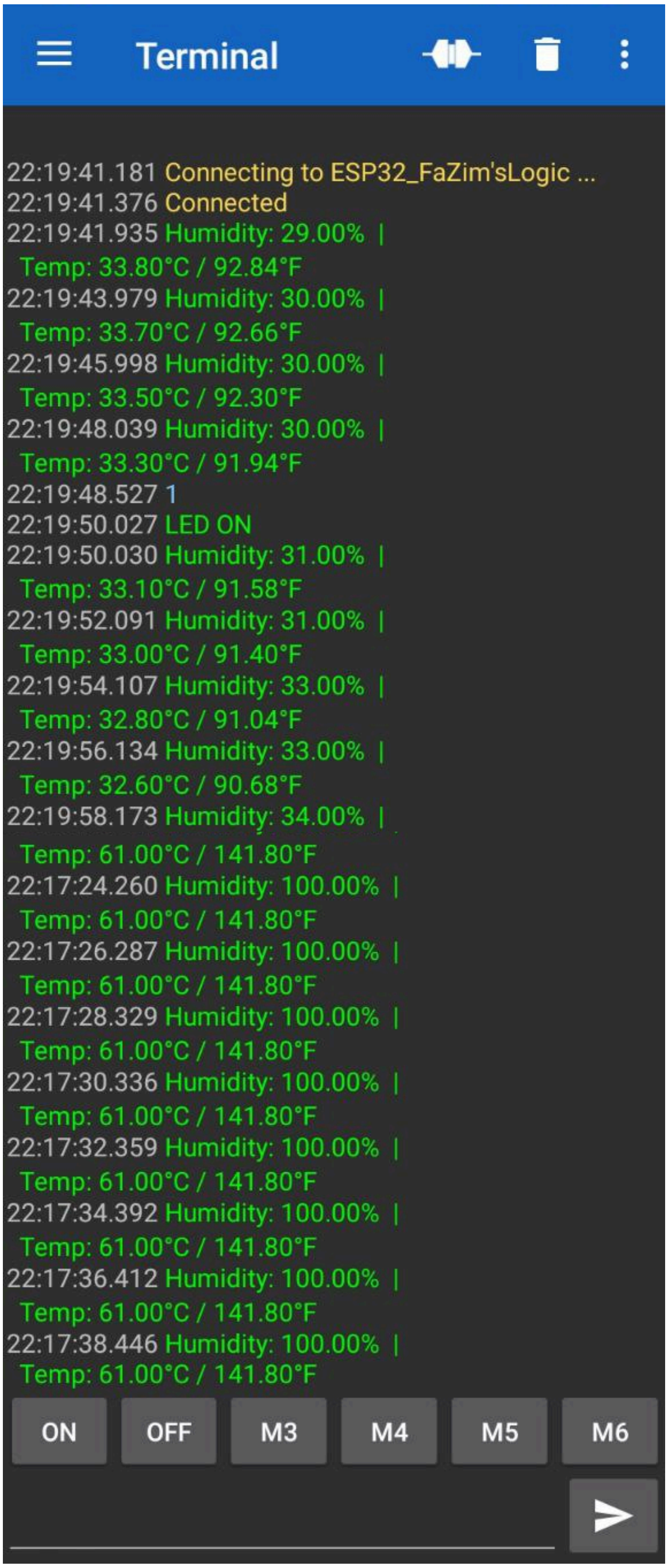

Figure 7.2 Shows the interface from Serial Bluetooth Terminal

8. Discussion

This experiment demonstrated how effectively the **ESP32** can integrate sensing, wireless communication, and basic actuation into a single system. Throughout the test, the microcontroller consistently read temperature and humidity data from the **DHT11 sensor** and transmitted it to the smartphone in real time. The stability of the readings over the 20–30 second sample period shows that the system was able to collect and update data reliably, even though the DHT11 is a simple, entry-level sensor. The slight changes in temperature and stable humidity values reflect normal behavior for indoor environmental conditions and confirm that the sensor was functioning as expected during operation.

Bluetooth communication played a central role in the system's performance, and its reliability became clear during testing. The **Serial Bluetooth Terminal app** received data without noticeable delay, allowing the user to monitor environmental changes as they occurred. The system also responded immediately when commands were sent from the smartphone to control the **LED**, demonstrating smooth two-way communication. This responsiveness indicates that Bluetooth is suitable for short-range real-time interactions in small-scale mechatronics projects, where immediate feedback and control are important.

Overall, the experiment successfully demonstrated the essential functions of a basic wireless monitoring and control system. It showed how environmental data can be captured, transmitted, and reacted to in real time using relatively simple hardware. The experience also emphasized the importance of reliable communication and sensor behavior when designing embedded systems. These results provide a solid understanding of how different components interact and lay the groundwork for more complex systems in future work.

9. Conclusion

In conclusion, this experiment successfully implemented a wireless temperature and humidity monitoring system with remote control capabilities using the **ESP32** microcontroller, **DHT11 sensor**, and LED indicator. The system consistently demonstrated its ability to collect real-time environmental data, send the information to a smartphone through **Bluetooth**, and **respond** immediately to user commands such as turning the LED ON or OFF.

These results confirm that the ESP32 is highly capable of integrating sensors, actuators, and wireless communication within a single platform, making it a practical tool for mechatronics and embedded system applications. Throughout the experiment, Bluetooth proved to be a stable and reliable method for short-range data transmission, offering smooth interaction between the microcontroller and smartphone without noticeable delays.

Beyond achieving all technical objectives, the experiment also provided valuable hands-on experience in essential areas such as microcontroller programming, digital sensor interfacing, serial communication, and remote actuation control. This practical exposure helps build a strong understanding of how modern IoT and smart device systems operate.

Overall, the experiment serves as a solid foundation for future enhancements, including advanced sensors, mobile app development, cloud-based monitoring, and larger-scale automation projects, demonstrating the potential of this simple system to evolve into a more sophisticated and real-world-ready smart monitoring solution.

10. Recommendations

Based on the experiment, the following recommendations can be made for improvement or future development:

1. **Upgrade the sensor:** Replace the DHT11 with DHT22 or BME280 for more accurate and faster readings. The DHT11 sensor is suitable for basic temperature and humidity measurements but has limitations in terms of accuracy and sampling speed. Replacing the DHT11 with a more advanced sensor, such as the **DHT22** or **BME280**, can provide more precise and faster readings. These sensors also offer a wider range of measurement, which is beneficial for applications requiring higher accuracy or more sensitive environmental monitoring.

2. **Actuator Expansion:** Use relays to control real appliances like fans, lamps, or heaters. In this experiment, the LED served as a simple actuator to simulate a FAN or indicator. For practical applications, **relays or other switching devices** can be added to control real appliances such as fans, lamps, or heaters. This would demonstrate how the system could be applied in home automation or industrial control scenarios, providing a real-world context for the experiment.

3. **Data Logging:** Store data to an SD card or cloud server for historical analysis. Currently, sensor readings are transmitted in real time but are not stored for later analysis. Adding **data logging capability** by saving data to an SD card or sending it to a cloud server would allow historical analysis, trend tracking, and performance evaluation over time. This can also facilitate predictive maintenance in practical applications, such as environmental monitoring or smart farming.

4. **Alert System:** Add alarms or notifications when temperature or humidity exceeds thresholds. Alarms, push notifications, or smartphone alerts can be triggered when temperature or humidity exceeds pre-defined thresholds. Such a feature is particularly useful for critical applications, such as server rooms, greenhouses, or laboratories, where maintaining specific environmental conditions is essential.

Implementing these recommendations will make the system more robust, versatile, and suitable for real-world applications.

11. References

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Appendix A - Circuit Diagram

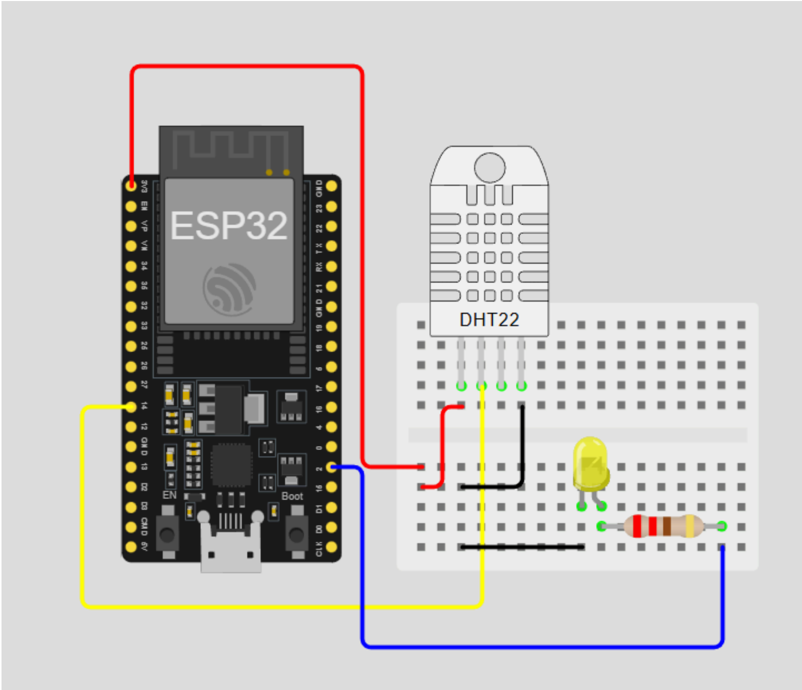

Figure A1 Shows the connection of circuit configuration by using Wokwi

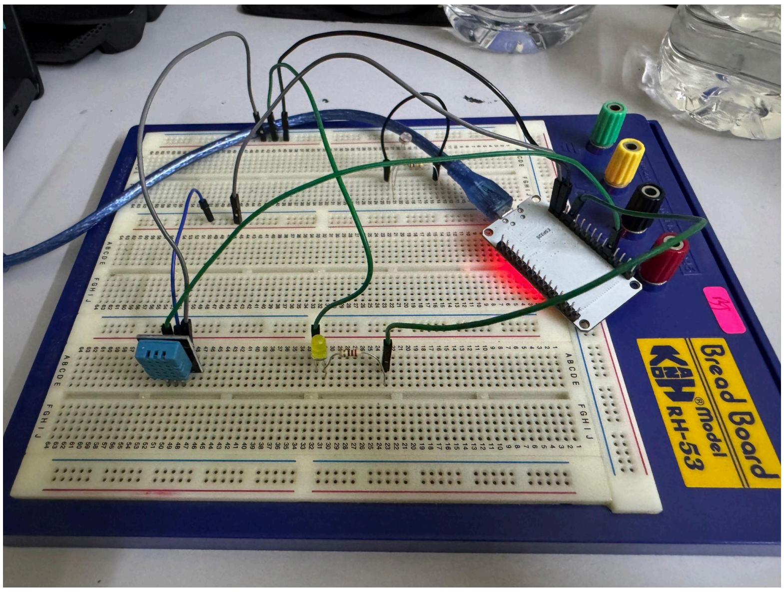

Figure A2 Shows the real connection of circuit hardware

Appendix B - Coding Snippets

```
#include <BluetoothSerial.h>
#include <DHT.h>

#define DHT11_PIN 21
#define DHTTYPE   DHT11
#define LED_PIN   2   // GPIO2 for external LED

String device_name = "ESP32_FaZim'sLogic";

BluetoothSerial SerialBT;
DHT dht11(DHT11_PIN, DHTTYPE);

#if !defined(CONFIG_BT_ENABLED) ||
!defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth not enabled in this ESP32
#endif

void setup() {
  Serial.begin(9600);
  SerialBT.begin(device_name);

  dht11.begin();

  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.println("ESP32 ready. Pair via Bluetooth.");
}

void loop() {
  // ---- Bluetooth LED Control ----
  if (SerialBT.available()) {
    char cmd = SerialBT.read();

    if (cmd == '1') {
      digitalWrite(LED_PIN, HIGH);
      SerialBT.println("LED ON");
    }
```

```
    else if (cmd == '0') {
      digitalWrite(LED_PIN, LOW);
      SerialBT.println("LED OFF");
    }
  }

  // ---- DHT11 readings ----
  float humi  = dht11.readHumidity();
  float tempC = dht11.readTemperature();
  float tempF = dht11.readTemperature(true);

  if (!isnan(humi) && !isnan(tempC)) {
    SerialBT.print("Humidity: ");
    SerialBT.print(humi);
    SerialBT.print("%  |  Temp: ");
    SerialBT.print(tempC);
    SerialBT.print("°C / ");
    SerialBT.print(tempF);
    SerialBT.println("°F");
  } else {
    SerialBT.println("Failed to read from DHT11");
  }

  delay(2000);
}
```

Appendix C - Results

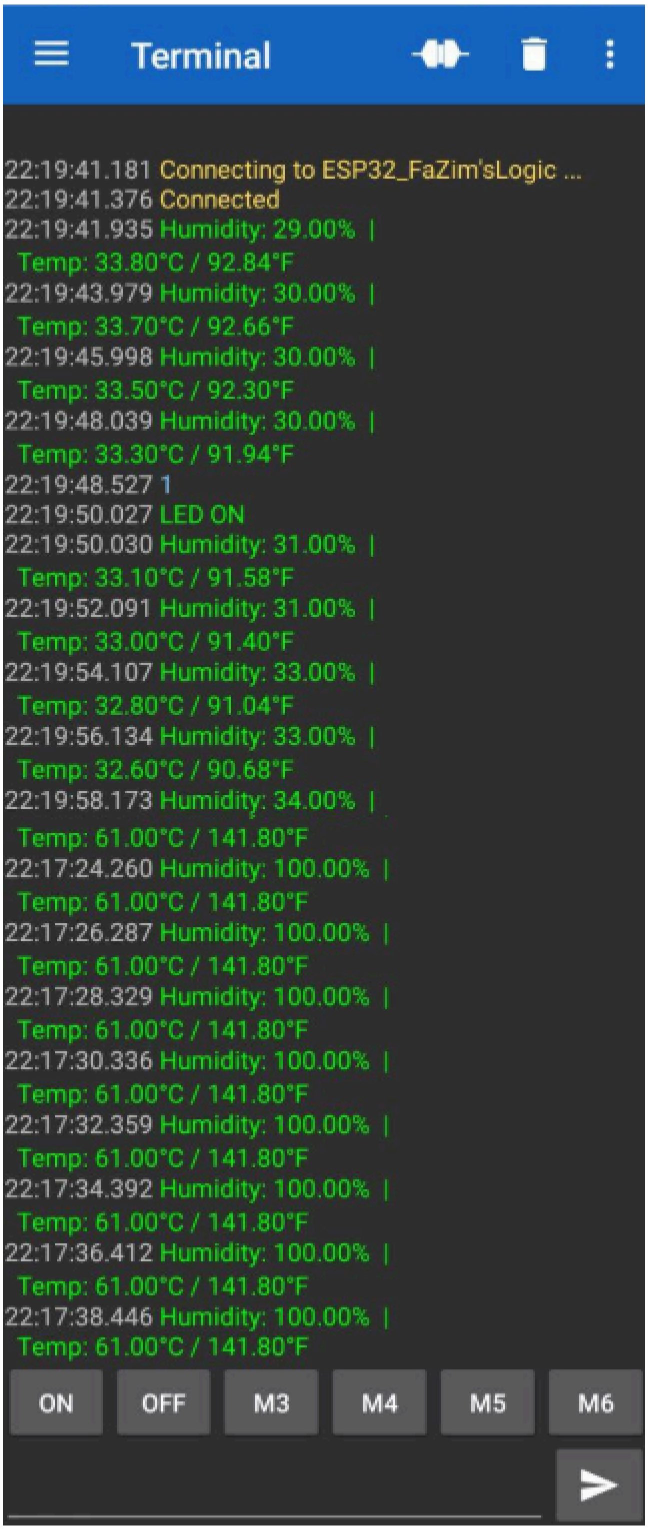

Figure C1 Shows the interface from Serial Bluetooth Terminal

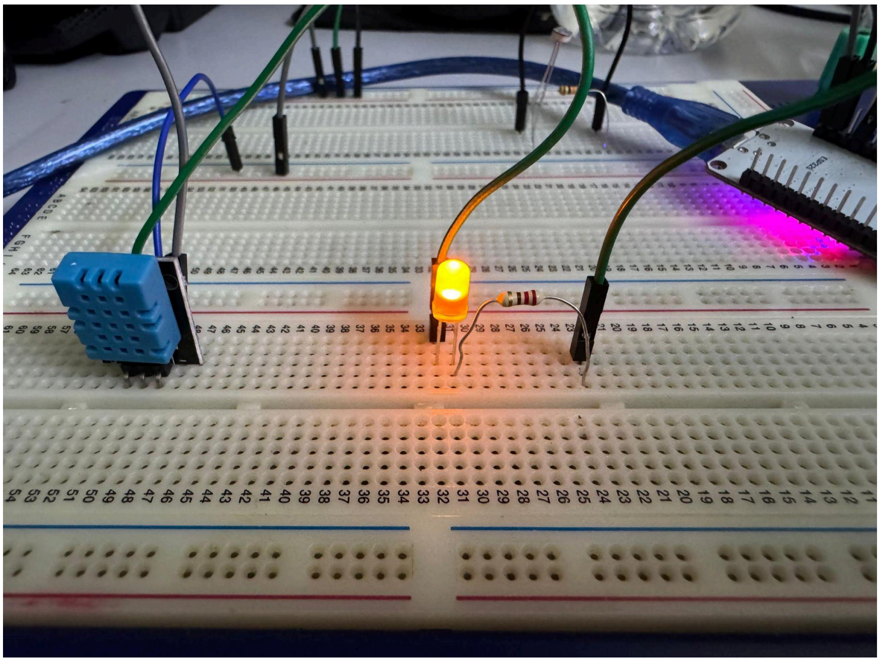

Figure C2 Shows the LED turn on after sending ‘1’ from Serial Bluetooth

Acknowledgments

We would like to express our sincere gratitude to all those who supported me during this laboratory experiment. First and foremost, I would like to thank my lecturer for providing guidance, clear instructions, and valuable advice throughout the experiment. Their explanations helped me understand how to integrate sensors, actuators, and Bluetooth communication with the ESP32.

I would also like to thank my classmates for their support and encouragement. Discussing ideas and troubleshooting problems together made the experiment easier and more enjoyable. Their help was especially valuable when setting up the hardware and debugging the code.

I am also grateful for the availability of the laboratory equipment, including the ESP32, DHT11 sensor, LED, breadboard, and jumper wires. Without these materials, it would not have been possible to complete the experiment successfully.

Finally, I would like to acknowledge my family and friends for their encouragement and patience during the preparation of this report. Their support gave me motivation to complete the experiment thoroughly and submit a complete lab report.

This experiment was a valuable learning experience, allowing me to develop practical skills in programming, sensor integration, and wireless communication, which will be useful in future mechatronics projects.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or a person.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

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Understand /

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